

HORNSBY GIRLS HIGH SCHOOL



Mathematics Extension 1

Year 11

Assessment 3 Term 3 2019

STUDENT NUMBER: _____

General Instructions

- Reading Time – 5 minutes
- Working Time – 80 minutes
- Write using black or blue pen
Black pen is preferred
- NESA-approved calculators and drawing templates may be used
- In Questions 6 – 9, show relevant mathematical reasoning and/or calculations
- Marks may be deducted for untidy and poorly arranged work
- Do not use correction fluid or tape
- Do not remove this paper from the examination

Total marks –

Section I Pages 2 – 3

5 marks

Attempt Questions 1 – 5

Answer on the Objective Response Answer Sheet provided

Section II Pages 5 – 14

47 marks

Attempt Questions 6 – 9.

Question	1-5	6	7	8	9	Total
Total	/5	/12	/11	/11	/13	/52

This assessment task constitutes 40% of the Preliminary Higher School Certificate Course School Assessment

Section I

5 marks

Attempt Questions 1 – 5

Allow about 10 minutes for this section

Use the Objective Response answer sheet for Questions 1 – 5

Q1. What is the derivative of $y = x^2 e^{2x}$?

(A) $y' = 4xe^{2x}$

(B) $y' = 2e^{2x}(1+x)$

(C) $y' = 2xe^{2x}(1+x)$

(D) $y' = 2xe^2$

Q2. Which of the following pairs of equations are parametric equations for $y = x^2 - 2x$?

(A) $x = t + 1, y = t^2 - 1$

(B) $x = t - 1, y = t^2 + 1$

(C) $x = t, y = t - 2$

(D) $x = t, y = t^2 - 2$

Q3. Simplify $\frac{1 + \tan 5x \tan 3x}{\tan 5x - \tan 3x}$

(A) $\cot 2x$

(B) $\tan 2x$

(C) $\frac{\tan 2x}{1 + \tan 5x \tan 3x}$

(D) $\cot x$

Q4. How many solutions does the equation $(\cot x - 1)(\cos x + 1) = 0$ have in the domain

$$0 \leq x \leq 2\pi ?$$

(A) 1

(B) 2

(C) 3

(D) 4

Q5. If $t = \tan \frac{\theta}{2}$ which of the following expressions is equivalent to $4 \sin \theta + 3 \cos \theta + 5$?

(A) $\frac{2(t+2)^2}{1-t^2}$

(B) $\frac{(t+4)^2}{1-t^2}$

(C) $\frac{2(t+2)^2}{1+t^2}$

(D) $\frac{(t+4)^2}{1+t^2}$

End of Section I

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Section II

47 marks

Attempt Questions 6-9

Allow about 70 minutes for this section

Question 6 (12 marks)

- (a) What is the remainder when $P(x) = 2x^4 - 5x^3 + 12x - 7$ is divided by $(x - 2)$? 1

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- (b) (i) Write down the expansion for $\cos(A + B)$. 1

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- (ii) Hence, or otherwise, find the exact value of $\sec 105^\circ$. Express your answer with a rational denominator. 2

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- (c) Solve for x : $\frac{x-5}{x+1} \geq 3$. 2

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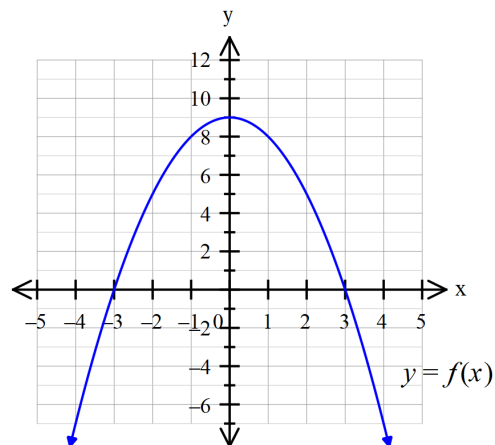
(d) State the exact value of $\cos^{-1}\left(\cos\frac{7\pi}{6}\right)$.

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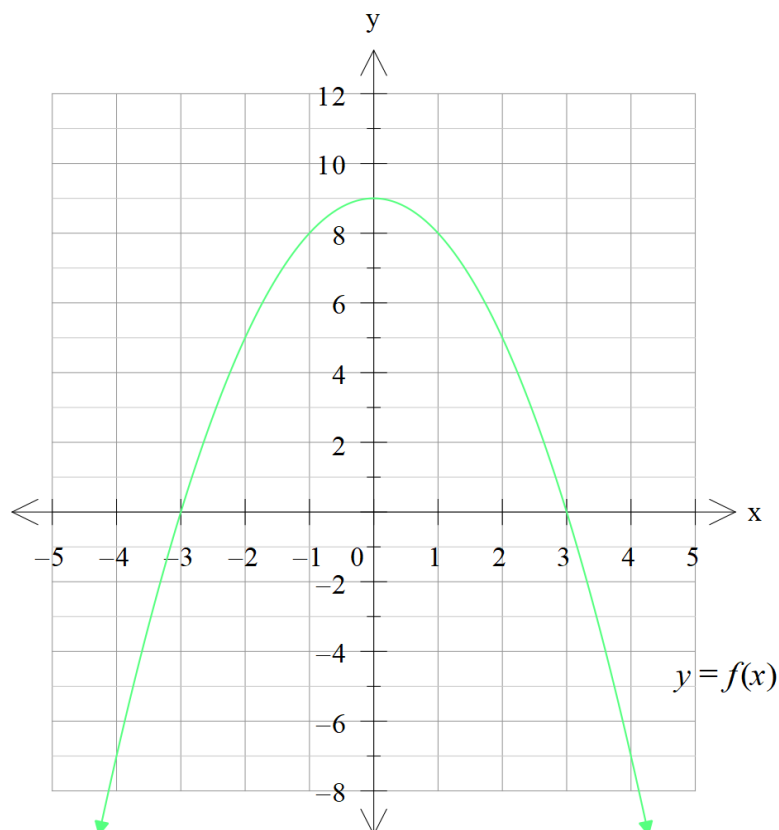
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(e) The graph $f(x) = 9 - x^2$ is drawn below.

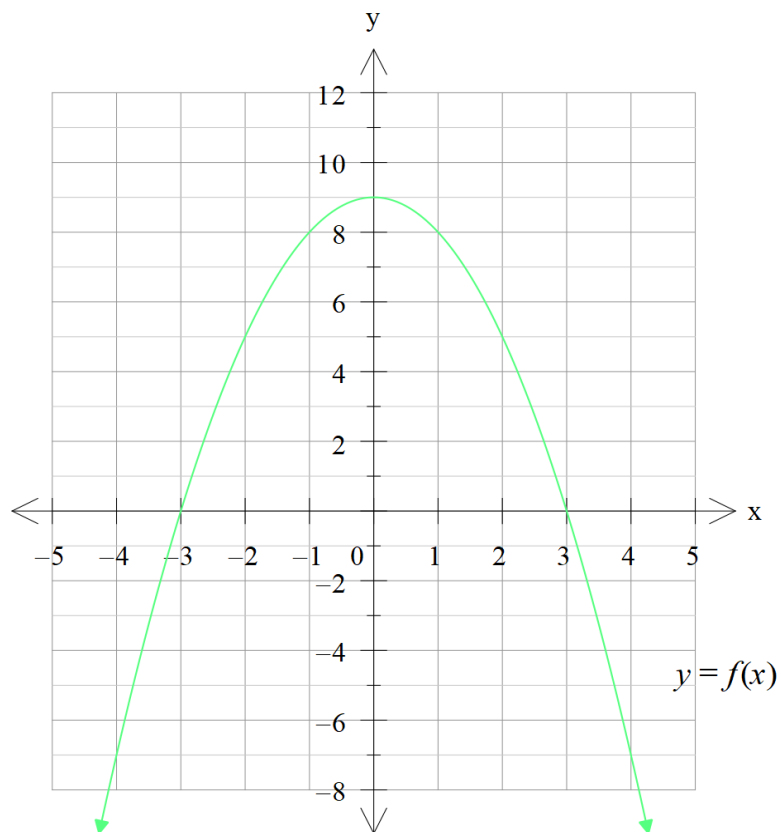


NOT TO SCALE

(i) On the graph below, sketch $y = \frac{1}{f(x)}$, showing all important features including intercepts with the coordinate axes and asymptotes. 2



- (ii) On the graph below, sketch $y = |f(x)| - 3$, showing all important features including intercepts with the coordinate axes and asymptotes. 2



Question 7 (11 marks)

- (a) If α, β and γ are the roots of $3x^3 - 6x^2 + 3x - 1 = 0$, find the value of **2**

$$\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$$

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- (b) Solve for θ , $0 \leq \theta \leq 2\pi$, $\cos 2\theta = \sin \theta$ **3**

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Question 7 continues on the next page

(c) A polynomial is defined as $P(x) = (x-2)^2(x+2)^2$

(i) Neatly sketch $y = P(x)$, clearly showing all intercepts with the coordinate axes.

2

(ii) Hence, or otherwise, find the value(s) of k for which $(x-2)^2(x+2)^2 = k$ has at least 3 solutions.

1

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(d) Let $f(x) = x^3 + kx^2 + 3x - 10$, where k is a constant. Find the values of k for which $f'(x) = 0$ has exactly two solutions.

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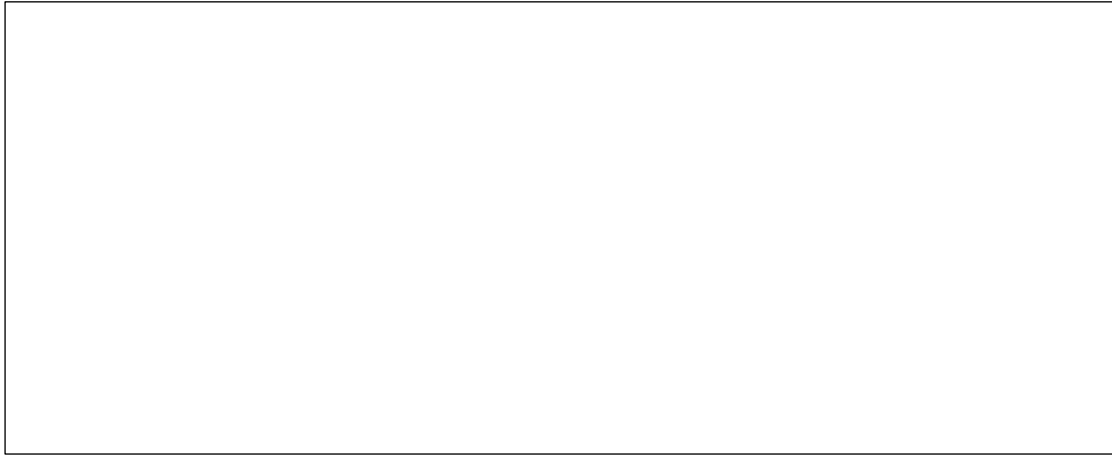
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Question 8 (11 marks)

(a) Sketch the function $y = 3 \cos^{-1} 2x$

2



(b) (i) Show that $(x+3)$ is a factor of $P(x) = x^3 - 5x + 12$

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(ii) Show that $(x+3)$ is the only linear factor of $P(x)$
over the field of Real Numbers for x .

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Question 8 continues on the next page

- (c) Find the exact value of $\sin\left(2 \tan^{-1}\left(\frac{3}{4}\right)\right)$ 2

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- (d) The rate of growth of a seal colony is proportional to the excess of the colony's population, N , over 500 and is given by $\frac{dN}{dt} = k(N - 500)$.

- (i) Show that $N = 500 + Ae^{kt}$, where t is the time in years, satisfies the 1
differential equation $\frac{dN}{dt} = k(N - 500)$.

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- (ii) The initial population of seals in the colony was 1 500. After five years, 2
the seal colony had doubled in size. Find the exact values of A and k .

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- (iii) Calculate the expected population of seals in the colony after 7 years. 1
Answer correct to the nearest seal.

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Question 9 (13 marks)

(a) Let $P(x) = (x+1)(x-3)Q(x) + a(x+1) + b$ where $Q(x)$ is a polynomial where a and b are real numbers.

When $P(x)$ is divided by $(x+1)$, the remainder is -11 .

When $P(x)$ is divided by $(x-3)$, the remainder is 1 .

(i) What is the value of b ?

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(ii) What is the remainder when $P(x)$ is divided by $(x+1)(x-3)$?

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Question 9 continues on the next page

(b) Use products to sums formulas, or otherwise, to solve

3

$$2(\sin \theta + 2 \cos 2\theta \sin \theta) = 1 \quad \text{for } 0 \leq \theta \leq \pi.$$

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(c) A spherical balloon is being deflated at a rate of $2\pi \text{ cm}^3/\text{s}$. Calculate the rate of change of its radius at the instant the radius is 5cm.

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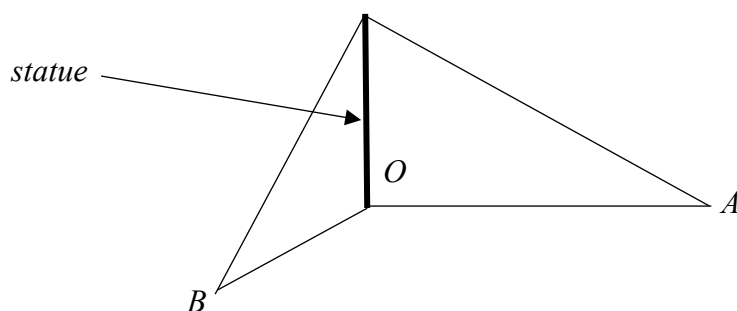
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Question 9 continues on the next page

Question 9 continued:

(d) Two tourists, A and B , are observing the top of an ancient statue. Tourist A is standing due east of the statue whereas Tourist B is standing on a bearing of 210° from the base of the statue. The angles of elevation of the top of the statue from Tourist A and Tourist B are 43° and 61° respectively. Let the base of the statue be O .



(i) Explain why $\angle AOB = 120^\circ$

1

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(ii) If h is the height of the statue and the distance between the tourists is 20 metres, show that:

3

$$h = \frac{20}{\sqrt{\tan^2 29^\circ + \tan^2 47^\circ + \tan 29^\circ \tan 47^\circ}}$$

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End of paper

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Mathematics Extension 1

Year 11 Assessment 3 Term 3 2019

STUDENT NUMBER:.....

Section I – Multiple Choice Answer Sheet

Allow about 10 minutes for this section

Select the alternative A, B, C or D that best answers the question. Fill in the response oval completely.

Sample: $2 + 4 =$ (A) 2 (B) 6 (C) 8 (D) 9
A ☐ B ☒ C ☐ D ☐

If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

A ☒ B ☒ C ☐ D ☐

If you change your mind and have crossed out what you consider to be the correct answer, then indicate the correct answer by writing the word **correct** and drawing an arrow as follows.

A ☒ B ☒ ^{correct} C ☐ D ☐

1. A ☐ B ☐ C ☐ D ☐

2. A ☐ B ☐ C ☐ D ☐

3. A ☐ B ☐ C ☐ D ☐

4. A ☐ B ☐ C ☐ D ☐

5. A ☐ B ☐ C ☐ D ☐

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Mathematics Extension 1

Year 11

Assessment 3 Term 3 2019

STUDENT NUMBER:

SOLUTIONS

General Instructions

- Reading Time – 5 minutes
- Working Time – 80 minutes
- Write using black or blue pen
Black pen is preferred
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- Marks may be deducted for untidy and poorly arranged work
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Total marks –

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5 marks

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Question	1-5	6	7	8	9	Total
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Section I

5 marks

Attempt Questions 1 – 5

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Use the Objective Response answer sheet for Questions 1 – 5

Q1. What is the derivative of $y = x^2 e^{2x}$?

(A) $y' = 4xe^{2x}$

(B) $y' = 2e^{2x}(1+x)$

(C) $y' = 2xe^{2x}(1+x)$

(D) $y' = 2xe^2$

$$y' = x^2 \cdot 2e^{2x} + 2xe^{2x}$$
$$y' = 2xe^{2x}(x+1)$$

Q2. Which of the following pairs of equations are parametric equations for $y = x^2 - 2x$?

(A) $x = t+1, y = t^2 - 1$

(B) $x = t-1, y = t^2 + 1$

(C) $x = t, y = t-2$

(D) $x = t, y = t^2 - 2$

$$(t+1)^2 - 2(t+1)$$
$$= t^2 + 2t + 1 - 2t - 2$$
$$= t^2 - 1$$

Q3. Simplify $\frac{1 + \tan 5x \tan 3x}{\tan 5x - \tan 3x}$

(A) $\cot 2x$

(B) $\tan 2x$

(C) $\frac{\tan 2x}{1 + \tan 5x \tan 3x}$

(D) $\cot x$

$$= \frac{1}{\frac{\tan 5x - \tan 3x}{1 + \tan 5x \tan 3x}}$$
$$= \frac{1}{\tan(5x - 3x)}$$
$$= \frac{1}{\tan 2x}$$
$$= \cot 2x$$

Q4. How many solutions does the equation $(\cot x - 1)(\cos x + 1) = 0$ have in the domain

$$0 \leq x \leq 2\pi?$$

(A) 1

(B) 2

(C) 3

(D) 4

$$\cot x = 1, \cos x = -1$$

$$\tan x = 1 \quad x = \pi$$

$$x = \frac{\pi}{4}, \frac{5\pi}{4}, \pi$$

but $x = \pi$ is undefined for $\cot x$

$\therefore$ only 2 solutions

Q5. If $t = \tan \frac{\theta}{2}$ which of the following expressions is equivalent to $4\sin \theta + 3\cos \theta + 5$?

(A) $\frac{2(t+2)^2}{1-t^2}$

(B) $\frac{(t+4)^2}{1-t^2}$

(C) $\frac{2(t+2)^2}{1+t^2}$

(D) $\frac{(t+4)^2}{1+t^2}$

$$4\left(\frac{2t}{1+t^2}\right) + 3\left(\frac{1-t^2}{1+t^2}\right) + 5$$

$$= \frac{8t + 3 - 3t^2 + 5 + 5t^2}{1+t^2}$$

$$= \frac{2t^2 + 8t + 8}{1+t^2}$$

$$= \frac{2(t^2 + 4t + 4)}{1+t^2}$$

$$= \frac{2(t+2)^2}{1+t^2}$$

End of Section I

Section II

47 marks

Attempt Questions 6-9

Allow about 70 minutes for this section

Question 6 (12 marks)

- (a) What is the remainder when $P(x) = 2x^4 - 5x^3 + 12x - 7$ is divided by $(x - 2)$? 1

$$P(2) = 2(2)^4 - 5(2)^3 + 12(2) - 7$$

$$= 9$$

- (b) (i) Write down the expansion for $\cos(A + B)$. 1

$$\cos(A + B) = \cos A \cos B - \sin A \sin B$$

- (ii) Hence, or otherwise, find the exact value of $\sec 105^\circ$. 2

$$\sec 105^\circ = \frac{1}{\cos 105^\circ} = \frac{1}{\cos(60^\circ + 45^\circ)}$$

many students made

technical error

of $\cos 60^\circ \cos 45^\circ - \sin 60^\circ \sin 45^\circ$

$$= \frac{\cos 60^\circ \cos 45^\circ - \sin 60^\circ \sin 45^\circ}{1}$$

$$= \frac{\frac{1}{2} \times \frac{\sqrt{2}}{2} - \frac{\sqrt{3}}{2} \cdot \frac{\sqrt{2}}{2}}{1}$$

$$= \frac{\frac{\sqrt{2}}{4} - \frac{\sqrt{6}}{4}}{1}$$

$$= \frac{\sqrt{2} - \sqrt{6}}{4}$$

$$= \frac{1}{\frac{\sqrt{2} - \sqrt{6}}{4}} = \frac{4}{\sqrt{2} - \sqrt{6}} = \frac{-4(\sqrt{2} + \sqrt{6})}{2 - 6} = -(\sqrt{2} + \sqrt{6})$$

- (c) Solve for x : $\frac{x-5}{x+1} \geq 3$. 2

$$\text{CV: } x \neq -1$$

$$\text{Consider: } \frac{x-5}{x+1} = 3$$

$$x-5 = 3x+3$$

$$-8 = 2x$$

$$\therefore x = -4$$

$$\text{Test: } x = -5, \frac{-10}{-4} \geq 3 \times$$

$$x = -2, \frac{-7}{-1} \geq 3 \checkmark$$

$$x = 0, \frac{-5}{1} \geq 3 \times$$

$$\therefore -4 \leq x < 1$$

Many multiplied both sides with perfect square then got lost in simplifying the inequality.

Many forgot to exclude $x = -1$.

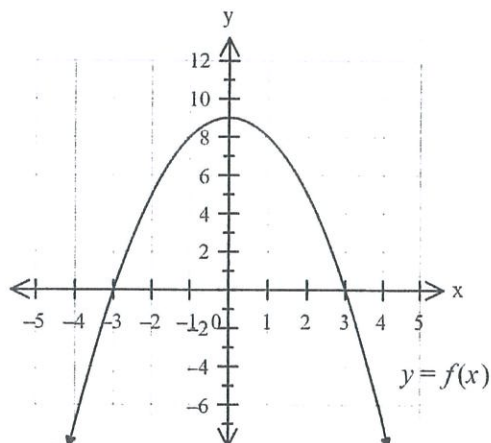
(d) State the exact value of $\cos^{-1}\left(\cos\frac{7\pi}{6}\right)$.

2

$$\cos\frac{7\pi}{6} = \cos\frac{5\pi}{6} \quad \therefore \cos^{-1}\left(\cos\frac{5\pi}{6}\right) = \frac{5\pi}{6}$$

many did not understand and how the domain applies.

(e) The graph $f(x) = 9 - x^2$ is drawn below.

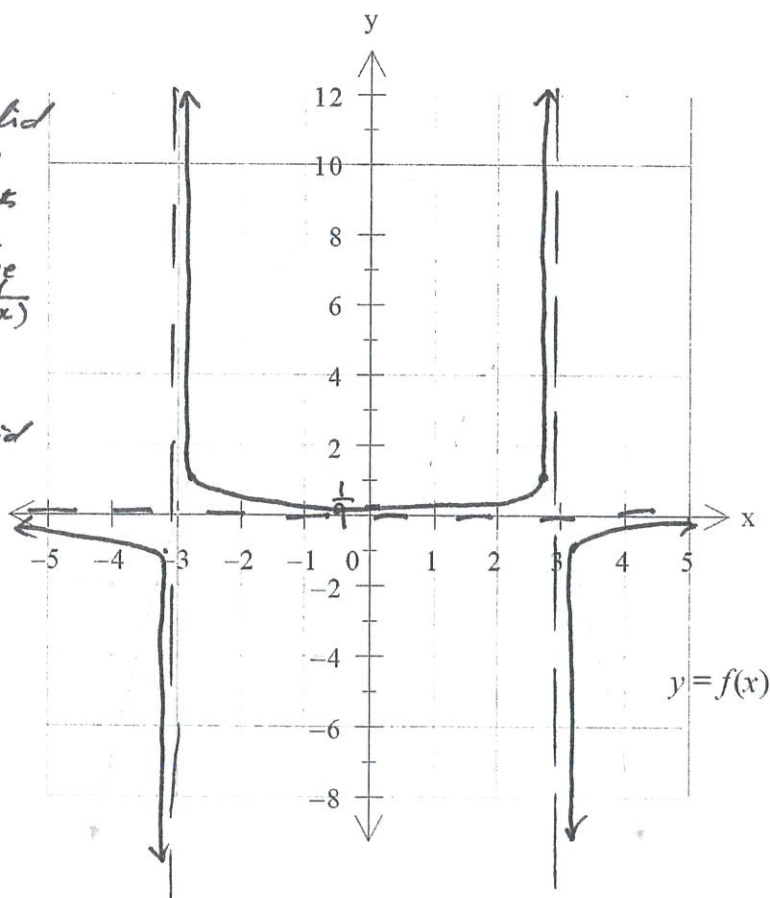


NOT TO SCALE

(i) On the graph below, sketch $y = \frac{1}{f(x)}$, showing all important

2

features including intercepts with the coordinate axes and asymptotes.

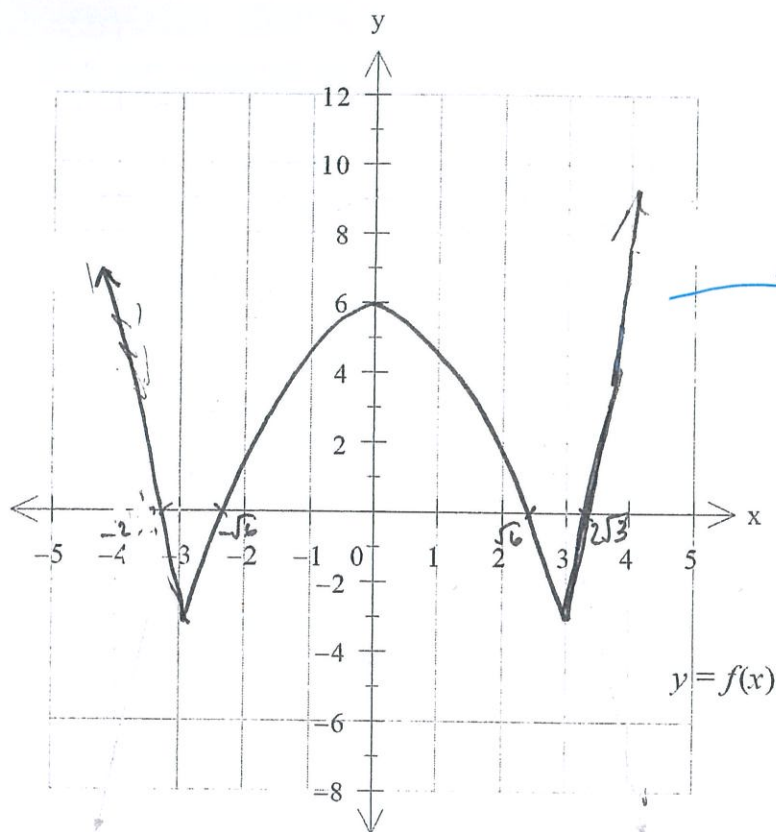


1) Many students did not pay attention to the fact that the points at line $y = \pm 1$ stay the same, hence the shape of the graph of $y = f(x)$ will not be sharp. is broader

2) Many students did not notice that as $x \rightarrow \pm\infty$, $y \rightarrow 0$, hence the x-axis is the horizontal asymptote.

- (ii) On the graph below, sketch $y = |f(x)| - 3$, showing all important features including intercepts with the coordinate axes and asymptotes.

2



→ may have the arms out.

$$y = |9 - x^2| - 3$$

$$0 = 9 - x^2 - 3 \quad \text{or} \quad 0 = -9 + x^2 - 3 \quad \text{for } x\text{-intercepts}$$

$$0 = 6 - x^2$$

$$\therefore x = \pm \sqrt{6}$$

$$0 = x^2 - 12$$

$$x^2 = 12$$

$$\therefore x = \pm 2\sqrt{3}$$

$$\therefore x\text{-int. are: } x = \pm \sqrt{6}, x = \pm 2\sqrt{3}$$

$$y = |9 - 0| - 3$$

$$y = 6$$

$$\therefore y\text{-int: } y = 6$$

Question 7 (11 marks)

(a) If α, β and γ are the roots of $3x^3 - 6x^2 + 3x - 1 = 0$, find the value of

2

$$\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$$

$$\begin{aligned} & \frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} \\ &= \frac{\beta\gamma + \alpha\gamma + \alpha\beta}{\alpha\beta\gamma} \end{aligned}$$

$$\alpha\beta + \beta\gamma + \gamma\alpha = 1$$

$$\alpha\beta\gamma = \frac{1}{3}$$

$$= \frac{1}{\frac{1}{3}}$$

$$= 3$$

Generally well done.

However some students do not remember the connection b/w the roots & the coefficients.

(b) Solve for θ , $0 \leq \theta \leq 2\pi$, $\cos 2\theta = \sin \theta$

3

$$1 - 2\sin^2 \theta = \sin \theta$$

$$2\sin^2 \theta + \sin \theta - 1 = 0$$

$$(2\sin \theta - 1)(\sin \theta + 1) = 0$$

$$\therefore \sin \theta = \frac{1}{2}, \sin \theta = -1$$

$$\therefore \theta = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{3\pi}{2}$$

$$\text{ref. } \angle = \frac{\pi}{6}$$

Partly done.

Some students wrote

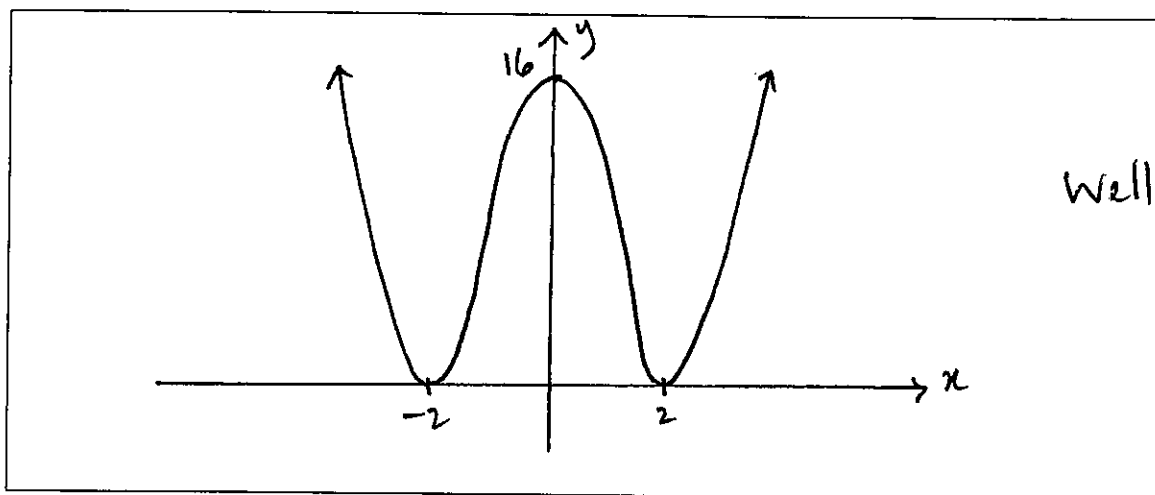
$\cos 2\theta$ as $1 - \sin^2 \theta$. Others wrote $\cos 2\theta = \cos(90 - \theta)$ & only solved for acute values of θ . Some students forgot to convert the angles from degrees to radians.

Question 7 continues on the next page

(c) A polynomial is defined as $P(x) = (x-2)^2(x+2)^2$

(i) Neatly sketch $y = P(x)$, clearly showing all intercepts with the coordinate axes.

2



Well done!

(ii) Hence, or otherwise, find the value(s) of k for which $(x-2)^2(x+2)^2 = k$ has at least 3 solutions.

1

Poorly done! Students did not make the connection of $y=k$ on the above graph.

(e) Let $f(x) = x^3 + kx^2 + 3x - 10$, where k is a constant. Find the values of k for which $f'(x) = 0$.

$$f'(x) = 3x^2 + 2kx + 3$$

$$0 = 3x^2 + 2kx + 3$$

$$\Delta = 4k^2 - 4(3)(3) > 0 \text{ for two solutions}$$

$$4k^2 - 36 > 0$$

$$k^2 - 9 > 0$$

$$\therefore (k-3)(k+3) > 0$$

$$\therefore k < -3, k > 3$$

Poorly done. Very few students used $\Delta > 0$ for two solutions

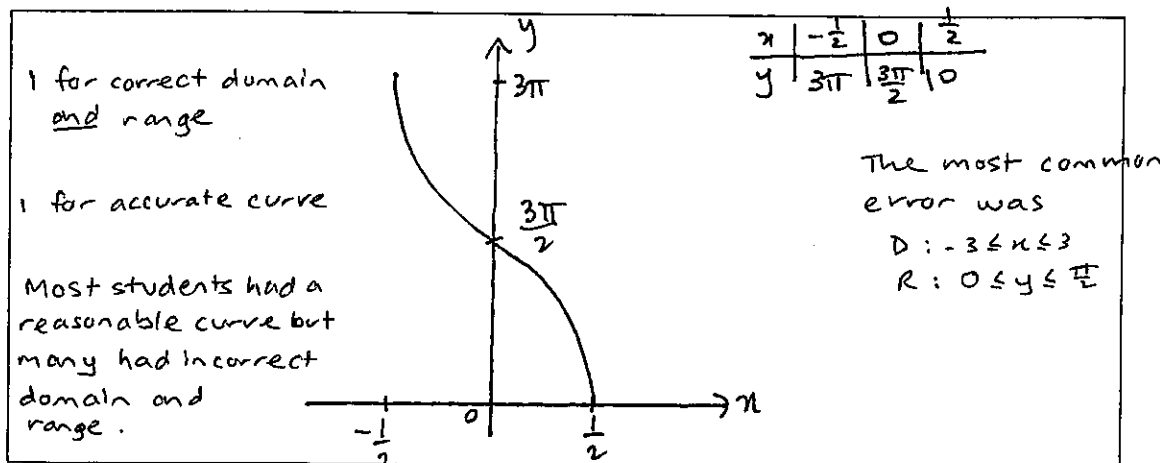
Question 8 (11 marks)

(a) Sketch the function $y = 3 \cos^{-1} 2x$

$$-1 \leq 2x \leq 1$$

$$\therefore -\frac{1}{2} \leq x \leq \frac{1}{2}$$

2



(b) (i) Show that $(x+3)$ is a factor of $P(x) = x^3 - 5x + 12$

1

$$P(-3) = (-3)^3 - 5(-3) + 12$$

$$P(-3) = -27 + 15 + 12$$

$$P(-3) = 0$$

Since $P(-3) = 0$, it follows that $(x+3)$ is a factor of $P(x)$.

(ii) Show that $(x+3)$ is the only linear factor of $P(x)$ over the field of Real Numbers for x .

2

$$\begin{array}{r} x^2 - 3x + 4 \\ x+3 \overline{) x^3 + 0x^2 - 5x + 12} \\ \underline{x^3 + 3x^2} \\ -3x^2 - 5x \\ \underline{-3x^2 - 9x} \\ 4x + 12 \\ \underline{4x + 12} \\ 0 \end{array}$$

$$\therefore P(x) = (x+3)(x^2 - 3x + 4)$$

for $x^2 - 3x + 4$, $\Delta = 9 - 4(4)$

$$\Delta = -7$$

$$\Delta < 0$$

$\therefore$ no real solutions

$\therefore (x+3)$ is the only real linear factor.

Question 8 continues on the next page

Some students checked all the integral zeros $\pm 1, \pm 2, \pm 3, \pm 4, \pm 6, \pm 12$ and found only $x = -3$ satisfies $P(x) = 0$ and then concluded that $(x+3)$ is the only linear factor. However, this does not consider the possibility of irrational (or complex) zeros.

e.g. $P(x) = x^3 + 2x^2 - 3x - 6$ could be checked for integral zeros at $\pm 1, \pm 2, \pm 3, \pm 6$ and -2 works. $P(x) \div (x+2)$ gives

$$P(x) = (x+2)(x^2 - 3)$$

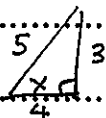
$\therefore$ the zeros are $-2, \pm \sqrt{3}$

(c) Find the exact value of $\sin\left(2\tan^{-1}\left(\frac{3}{4}\right)\right)$

2

let $x = \tan^{-1} \frac{3}{4}$

$\therefore \tan x = \frac{3}{4}$



Must show sufficient working for a 2 mark question.

$\therefore \sin 2x = 2 \sin x \cos x$

$= 2\left(\frac{3}{5}\right)\left(\frac{4}{5}\right)$

$= \frac{24}{25}$

correct answer with insufficient working was awarded one mark.

(c) The rate of growth of a seal colony is proportional to the excess of the colony's

population, N , over 500 and is given by $\frac{dN}{dt} = k(N - 500)$. Working must demonstrate that students know how to differentiate Ae^{kt}

(i) Show that $N = 500 + Ae^{kt}$, where t is the time in years, satisfies the 1

differential equation $\frac{dN}{dt} = k(N - 500)$. Very poor setting out by a large number of students.

$N = 500 + Ae^{kt}$

$N - 500 = Ae^{kt} \quad \text{--- ①}$

$\frac{dN}{dt} = kAe^{kt}$

$\therefore N = 500 + Ae^{kt}$ satisfies

$= k(N - 500)$ from ① the differential equation

(ii) The initial population of seals in the colony was 1 500. After five years, 2

the seal colony had doubled in size. Find the exact values of A and k .

when $t = 0$, $N = 1500$ $3000 = 500 + 1000e^{5k}$ Many students incorrectly found

$1500 = 500 + Ae^0$ $\frac{2500}{1000} = e^{5k}$ that $A = 1500$ and then

$\therefore A = 1000$ $\therefore 5k = \ln \frac{5}{2}$ $k = \frac{1}{5} \ln \frac{5}{2}$

$\therefore N = 500 + 1000e^{kt}$ $\therefore k = \frac{1}{5} \ln \frac{5}{2}$

when $t = 5$, $N = 3000$

(iii) Calculate the expected population of seals in the colony after 7 years. 1

Answer correct to the nearest seal.

$N = 500 + 1500e^{7\left(\frac{1}{5}\ln \frac{5}{2}\right)}$

which gave $N = 3567$

$N = 3567$ seals.

Question 9 (13 marks)

(a) Let $P(x) = (x+1)(x-3)Q(x) + a(x+1) + b$ where $Q(x)$ is a polynomial where a and b are real numbers.

When $P(x)$ is divided by $(x+1)$, the remainder is -11 .

When $P(x)$ is divided by $(x-3)$, the remainder is 1 .

(i) What is the value of b ?

1

$$-11 = 0 + a(0) + b$$

$$\therefore b = -11$$

(ii) What is the remainder when $P(x)$ is divided by $(x+1)(x-3)$?

2

$$\therefore P(x) = (x+1)(x-3)Q(x) + a(x+1) - 11$$

$$\text{when } x=3, P(3)=1$$

$$\therefore 1 = 0 + a(3+1) - 11$$

$$1 = 0 + 4a - 11$$

$$12 = 4a$$

$$\therefore a = 3$$

1 mark

$$\therefore \text{remainder is } 3(x+1) - 11$$

$$= 3x - 8$$

1 mark

Comment: many worked out a , but not able to see the remainder being $a(x+1) + b$.

Question 9 continues on the next page

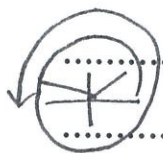
(b) Use products to sums formulas, or otherwise, to solve

3

$$2(\sin \theta + 2 \cos 2\theta \sin \theta) = 1 \quad \text{for } 0 \leq \theta \leq \pi.$$

$$2 \cos 2\theta \sin \theta = \sin 3\theta - \sin \theta \quad 0 \leq \theta \leq \pi$$

$$\therefore 2(\sin \theta + \sin 3\theta - \sin \theta) = 1 \quad (1 \text{ mark})$$



$$2 \sin 3\theta = 1$$

$$\therefore \sin 3\theta = \frac{1}{2} \quad (1 \text{ mark})$$

$$\therefore 0 \leq 3\theta \leq 3\pi$$

$$3\theta = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{13\pi}{6}, \frac{17\pi}{6}$$

$$\therefore \theta = \frac{\pi}{18}, \frac{5\pi}{18}, \frac{13\pi}{18}, \frac{17\pi}{18} \quad (1 \text{ mark})$$

Comment: * many couldn't see the identity

$$2 \cos 2\theta \sin \theta = \sin(2\theta + \theta) - \sin(2\theta - \theta) \\ = \sin 3\theta - \sin \theta$$

* many didn't check the solutions on the domain.

(c) A spherical balloon is being deflated at a rate of $2\pi \text{ cm}^3/\text{s}$. Calculate the rate of change of its radius at the instant the radius is 5cm.

3

$$\frac{dv}{dt} = -2\pi$$

$$V = \frac{4}{3}\pi r^3$$

$$\frac{dv}{dr} = 4\pi r^2 \quad (1 \text{ mark})$$

$$\frac{dv}{dr} = \frac{dv}{dt} \times \frac{dt}{dr}$$

$$\text{or: } \frac{dr}{dt} = \frac{dr}{dv} \times \frac{dv}{dt}$$

$$4\pi r^2 = -2\pi \times \frac{dt}{dr} \quad (1 \text{ mark})$$

$$\text{when } r = 5,$$

$$4\pi(5)^2 = -2\pi \times \frac{dt}{dr}$$

$$\frac{100\pi}{-2\pi} = \frac{dt}{dr}$$

$\therefore$ the radius is decreasing

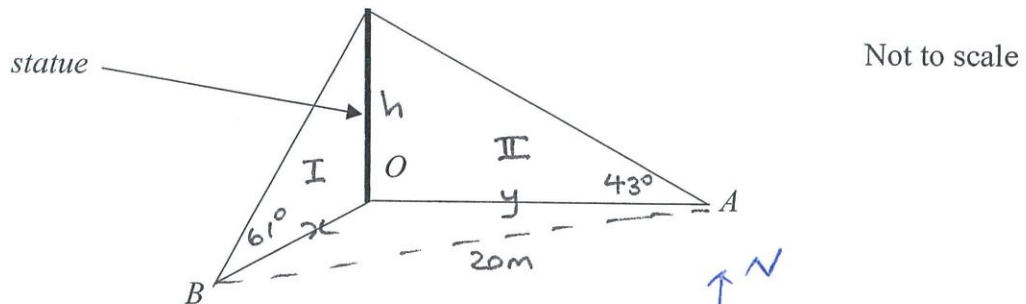
$$\therefore \frac{dr}{dt} = -\frac{1}{50} \quad \text{at a rate of } \frac{1}{50} \text{ cm/s.}$$

Question 9 continues on the next page

Comment: the answer should state the rate being decreasing or state $-\frac{1}{50} \text{ cm/s}$ in the answer.

Question 9 continued:

(d) Two tourists, A and B , are observing the top of an ancient statue. Tourist A is standing due east of the statue whereas Tourist B is standing on a bearing of 210° from the base of the statue. The angles of elevation of the top of the statue from Tourist A and Tourist B are 43° and 61° respectively. Let the base of the statue be O .



(i) Explain why $\angle AOB = 120^\circ$

$$\angle AOB = 90^\circ + 30^\circ$$

$$= 120^\circ$$

Or $210^\circ - 90^\circ$

(ii) If h is the height of the statue and the distance between the tourists

is 20 metres, show that:

$$h = \frac{20}{\sqrt{\tan^2 29^\circ + \tan^2 47^\circ + \tan 29^\circ \tan 47^\circ}}$$

In ΔI , $\tan 61^\circ = \frac{h}{x}$

$$\therefore x = \frac{h}{\tan 61^\circ}$$

$$x = h \cot 61^\circ$$

$$\therefore x = h \tan 29^\circ$$

Similarly, in ΔII , $y = h \tan 47^\circ$

$$\begin{aligned}
 \therefore 20^2 &= (h \tan 29^\circ)^2 + (h \tan 47^\circ)^2 - 2(h \tan 29^\circ)(h \tan 47^\circ) \cos 120^\circ \\
 20^2 &= h^2 \tan^2 29^\circ + h^2 \tan^2 47^\circ - 2h^2 \tan 29^\circ \tan 47^\circ \left(-\frac{1}{2}\right) \\
 20^2 &= h^2 \tan^2 29^\circ + h^2 \tan^2 47^\circ + h^2 \tan 29^\circ \tan 47^\circ \\
 20^2 &= h^2 [\tan^2 29^\circ + \tan^2 47^\circ + \tan 29^\circ \tan 47^\circ] \\
 \therefore h^2 &= \frac{20^2}{\tan^2 29^\circ + \tan^2 47^\circ + \tan 29^\circ \tan 47^\circ} \\
 h &= \frac{20}{\sqrt{\tan^2 29^\circ + \tan^2 47^\circ + \tan 29^\circ \tan 47^\circ}}, \quad h > 0
 \end{aligned}$$

comment = need to show $\cos 120^\circ = -\frac{1}{2}$
 - ~~very difficult~~
 need to show $\tan \theta = \tan(90^\circ - \theta)$
 or work out the
 tan ratio for the
 complement angle. 

End of paper